

Nicolas Giordano

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Passionate scientist motivated by bridging state-of-the-art data analytics techniques and crop eco-physiology principles to boost on-farm productivity and narrow yield gaps. Dedicated to continuous process improvement with a strong track record of promoting cross-functional collaboration and delivering actionable outcomes.

Education

Kansas State University **Manhattan, KS**
Ph.D., Agronomy
May 2025

Dissertation: “Strategies for sustainable improvement of the wheat phenotype”.

Kansas State University **Manhattan, KS**
Graduate Certificate, Applied Statistics
May 2024

University of Buenos Aires **Buenos Aires, Argentina**
B.S., Agricultural Engineering
March 2020

Thesis: “Effects of foliar fungicides timing on the incidence of late-season diseases in soybean”.

Professional Experience

University of Nebraska-Lincoln, Department of Agronomy and Horticulture **Lincoln, NE**
Postdoctoral Research Associate March 2026- Present

- Leading project exploring nitrogen fertilization strategies, and agronomic management, and their impact on environmental outcomes of in the US corn belt.
- Leading a global assessment on labor productivity across major upland and lowland crops.
- Collaborating with stakeholders from all five continents, to establish a global network of yield potential experiments in major upland and lowland crops.

Kansas State University, Department of Agronomy **Manhattan, KS**
Postdoctoral Fellow May 2025- February 2026

- Awarded with the Natural Resources and Environmental Sciences Fellowship, securing \$2,000 to guide a team of students to develop a Python pipeline that will assist sorting of seeds used in field trials.
- Worked on project exploring magnitudes and causes of wheat yield gaps in North America.
- Leded project on wheat value chain sustainability.
- Mentored fellow graduate student on their research projects.
- Collaborated on project that uses life-cycle-assessment to explore high productivity-low emissions crops.

Kansas State University, Department of Agronomy **Manhattan, KS**
Graduate Research Assistant January 2021 – May 2025

- Developed machine learning and Bayesian models to predict impacts of genotypes, management and environment on crop yield and quality traits.
- Demonstrated proficiency in handling, analyzing, and visualizing data using programming software (*e.g.*, R, Python, Stan, ArcGIS Pro).
- Developed programming and conceptual pipelines to integrate environment and genomic data.
- Collaborated with computer scientists and plant pathologists to build knowledge graphs on crop disease and nitrogen management in wheat.

- Developed *IDataField*, a Python-based platform for planning, monitoring, collecting, and processing data from field trials.
- Built Shiny and Streamlit applications for end-user interaction with research outcomes.
- Secured a \$25,000 grant for the research project “Genetic modulation of yield and yield plasticity in winter wheat”. Developed research objectives, study rationale, and budget.
- Led and collaborated on multiple research projects focused on genetic, environmental, and agronomic factors affecting crop grain yield and quality at regional and global scales.
- Published research in high-impact scientific journals.
- Delivered four oral presentations in international scientific conferences, including one at invited symposium.
- Delivered eight extension talks in crop schools and field days to approximately 250 growers and consultants across Kansas.
- Led the Wheat Value Chain Sustainability Summit, coordinating an expert elicitation workshop where 46 industry stakeholders defined sustainability across the wheat industry.
- Coordinated and executed field operations and trials across ~10 locations over four years, guiding field crews and mentoring summer interns.
- Served as teaching assistant in Crop Science (AGRON 220) and guest lecturer in World Food Crops (AGRON 325).
- Provided statistical consulting to animal science specialists in Argentina, Chile, and the US.
- Mentored two undergraduate students in data analysis and thesis writing.
- Conducted quantitative literature reviews, resulting in a high-impact publication and a research collaboration with plant pathology experts.
- Delivered a lecture on quantitative literature reviews using R for agronomy graduate students.
- Led and organized weekly meetings for the KSU Wheat and Forages Lab during three-years.
- Delivered lectures for interns and visiting scholars in quantitative methods for agricultural science, during one semester.

South Australian Research and Development Institute

Short Term Assignment

Adelaide, Australia

September 2024 – November 2024

- Collaboration with Australian scientists and remotely with 16 other scientists from diverse backgrounds (breeders, geneticists, crop physiologists, agronomists, and plant pathologists) across four different countries on a project to investigate the genetic modulation of yield phenotypic plasticity in winter wheat genotypes.
- Performed data analysis, visualization, and write-up of the research project.
- Published a manuscript from this work in the *Journal of Experimental Botany*.

Kume Mapu S.A.

Grower

Lincoln, Argentina

March 2014 – December 2020

- Gained extensive experience working at my family farming company, contributing to multiple operations from row crop farming to cow-calf, feedlot, and dairy farm operations.
- Contributed to various roles, including harvest logistics, planting, crop scouting, range and cattle management, commodity commercialization, investment valuations, and rural employee management.
- Coordinated on-farm research trials in 2020, collaborating with seed dealers and a growers’ group.
- Redesigned crop rotations and field management zones using remote sensing and field history records.
- Timely coordination and execution of summer row crop harvest and winter crops planting.

Pedro A. Lacau e Hijos S.R.L.

Field Agronomist Intern

Arenaza, Argentina

December-March 2015-2020

- Gained experience in soybeans, corn, wheat, and sunflower management during summer internships at a farming company.

- Roles included scouting over 8,000 acres each season, coordinating harvest logistics, informing insecticide and fungicide recommendations.
- Executed on-farm research trials in three locations in the 2017 harvest season.
- Collected data from on-farm trials for my undergraduate thesis project.

ALFIN S.C.A.

Field Agronomist Intern

Chaco, Argentina

July 2019

- Winter internship at a farming operation. Responsible for logistics, supervision, and timely execution of maize harvest on a 7,600-acre farm located in the northeastern region of Argentina.

University of Kentucky

Visiting Scholar

Lexington, KY

April 2018 -November 2018

- Collected data from field trials from a soft winter wheat breeding program.
- Performed disease rating of Fusarium head blight.
- Set up greenhouse experiments, performed field husbandry operations, processed laboratory samples, and data entry.

Peer-reviewed publications (under review 2/in preparation 4)

Fahmuddin A.; Pratamaningsih M.M.; Hati D.P.; Erwinda E.; Muslim R.Q.; Purwantomo D. K. G.; Mulyani A.; **Giordano N.**; Aramburu-Merlos F., Grassini P. Farmland protection and yield intensification are needed to avoid large rice imports and land conversion in Indonesia (2026). Land Use Policy. Under review.

Giordano, N., Sadras, V.O., Griffin, T., Nelson, K. S., Gautam, A., Welch, S., Guttieri, M., McGinty, H. K., Gelal, N., Bonini Pires, C., Dalal, A. S., Andersen Onofre, K., Ribert Francois, J., Kohl, D., Hefley, T., Hitzler, P., & Lollato, R. P. (2026). Fragmented consensus: Industry challenges in defining sustainable wheat. PNAS. Under review.

Romero Soler, J., Grassini P., Broyaka A., Pilecco, I., **Giordano N.**, Andersen Onofre K., Draper M.A., Shanoyan A., Featherstone A., Romulo P. Lollato R.P. (2026). Narrowing wheat yield gaps in Armenia and Georgia for food security in times of war. Agricultural systems. Under review.

Lollato R.P., B.L. Beres, J. Andrade, J.A. Romero Soler, Z. Wang, **N. Giordano**, A. Couëdel, I. Pilecco, F.A. Merlos, P. Grassini. (2026). Increasing yields in North America to meet global wheat demand. Nature Food. 2025. In preparation.

Dias da Silva Jr. C., **N. Giordano**, R.P. Lollato, E. De Wolf, K.A. Andersen Onofre. (2026). Meta-analysis reveals environmental drivers of wheat grain yield response to fungicide application timing to control stripe rust. Plant Disease. In preparation.

Pradella, L.O., M.J. Guttieri, R.L. Veenstra Cott, **N. Giordano**, R.P. Lollato. (2025). Phenotypic plasticity of tillering modulates winter wheat agronomic optimum plant density: A quantitative framework for improved interpretation of G×E×M interactions. European Journal of Agronomy. Under review.

Jaenisch B.R*, **N. Giordano***, V.O. Sadras, N.O. Nelson, M. Guttieri, D. Ruiz Diaz, A.K. Fritz, and R.P. Lollato. (2025). Nutrient use efficiency and co-limitation for nitrogen and sulfur in bread winter wheat. Field Crops Research. Under review.

Gelal, N., Gautam, A., Norouzi, S. S., **Giordano, N.**, Silva Jr, C. D. D., Francois, J. R., ... & McGinty, H. K. (2025). Building Knowledge Graphs Towards a Global Food Systems Datahub. arXiv preprint arXiv:2502.19507. In preparation.

Peer-reviewed publications (published 7)

- Giordano, N.,** Sadras, V. O., Guttieri, M. J., Cruppe, G., Bai, G., Fritz, A., Bernardo, A., St. Amand, P., Abbate, P., Berger, A., Boehm, J., De Oliveira Silva, A., Jones-Diamond, S., Haley, S., Lingenfelter, J., Zhang, G., & Lollato, R. (2025). Genetic modulation of yield and phenotypic plasticity of yield in winter wheat. *Journal of Experimental Botany*, eraf478. doi.org/10.1093/jxb/eraf478.
- Giordano, N.,** Sadras, V. O., Guttieri, M. J., Hefley, T. J., Romero Soler, J. A., Neely, C., Edwards, J. T., de Oliveira Silva, A., Fritz, A. K., and Lollato, R. P. (2025). High yield stability and susceptibility to stripe rust associate with large yield gaps in rainfed winter wheat. *Field Crops Research*, 333, 110118. doi.org/10.1016/j.fcr.2025.110118
- Giordano N.,** D. Hayes, T.J. Hefley, J. Lacasa, B.L. Beres, L. Haag, and R.P. Lollato. 2024. A Bayesian framework to model variance of grain yield response to plant density. *Plant Methods*. doi.org/10.1186/s13007-025-01355-y
- Lollato R.P., L.O. Pradella, **N. Giordano**, L.P. Ryan, J.R. Soler, L.M. Simao, B.R. Jaenisch, and R. Horton. 2024. Winter wheat response to plant density in yield contest fields. *Crop Science*. doi.org/10.1002/csc2.21296.
- Giordano, N.,** V.O. Sadras, A.A. Correndo, and R.P. Lollato. 2024. Cultivar-specific phenotypic plasticity of yield and grain protein concentration in response to nitrogen in winter wheat. *Field Crops Research*, 306, p.109202. doi.org/10.1016/j.fcr.2023.109202.
- Giordano, N.,** V.O. Sadras, and R.P. Lollato. 2023. Late-season nitrogen application increases grain protein concentration and is neutral for yield in wheat. A global meta-analysis. *Field Crops Research*, 290, p. 108740. doi.org/10.1016/j.fcr.2022.108740.
- Sadras, V.O., **N. Giordano**, A.A. Correndo, M. Cossani, J.M. Ferreyra, O.P. Caviglia, J.A. Coulter, I.A. Ciampitti, and R.P. Lollato. 2022. Temperature-driven developmental modulation of yield response to nitrogen in wheat and maize. *Frontiers in Agronomy*. 4:903340. doi.org/10.3389/fagro.2022.903340.

Leadership and Service

- **Referee for peer-reviewed journals:** *Field Crops Research* (9), *Agricultural Systems* (1), *Journal of Productivity Analysis* (1), *Journal of Plant Nutrition and Soil Science* (1), and *BMC Plant Biology* (1).
- **ASA Applied Wheat Research Community**, *Chair elect* (2025) and *Chair* (2026).
- **CSSA C3- Crop Ecology, Management and Quality**, Poster Competition Judge (2025).
- **ASA-CSSA-SSA**, *Member*. August 2021 – Present.
- **Agronomy Graduate Students Assn.**, *Statistics Committee Chair*. January 2024 – December 2024.
- **Agronomy Graduate Students Assn.**, *Social Chair*. January 2022 – December 2022.
- **Jornaderos Agro**, *President*. February 2019 – December 2019. Led a young-student non-profit organization focused on creating opportunities for undergraduate students to visit farmers across the central Pampas region in Argentina.

Honors and Awards

- Natural Resources and Environmental Sciences Graduate Fellowship. Kansas State University. May 2025.
- Bernard E. and Estelle L. Foote Scholarship in Agriculture, Kansas State University. July 2024.
- 1st place in Data Hackathon Competition, NUE Conference, Stillwater OK. August 2023.
- 2nd place in Data Hackathon Competition, NUE Conference, Lincoln, NE. August 2022.

- 1st place in Poster Competition, NUE Conference, Stillwater OK. August 2023.
- 2nd place in Poster Competition, ASA-CSSA-SSSA Conference, Salt Lake City, UT. November 2021.
- Scharder/Massier Graduate Excellence Fund in Agronomy, Kansas State University. July 2021.

Skills

- Fluent in English and Spanish.
- GitHub for software version control and collaboration in programming and data pipelines.
- Proficient in R and Python programming languages.